Jieun Kim

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RESEARCH INTERESTS

Human–AI Complementarity, Human vs. Machine Information Processing, Learning from Incomplete Real-World Data, Trustworthy AI, Computational Neural Modeling

Understanding how humans and AI systems differ in the way they process information, especially when real-world data are incomplete or noisy, and using this understanding to build AI systems that can take on tasks currently only humans can do and that reliably complement human judgment.

EDUCATION

Korea University Mar 2021 – Feb 2023
M.S. in Industrial and Management Engineering
Advisor: Jun-Geol Baek

Sookmyung Women's University Mar 2017 – Feb 2021
B.S. in Statistics and Computer Science
Advisor: Young-Geun Choi

RESEARCH EXPERIENCE

Korea Institute of Science and Technology (KIST) May 2024 – Jan 2026
Research Intern (Advisor: Taegon Kim)

- Constructed large-scale spiking neural network (SNN) models of cortico-basal ganglia circuits to study how biological networks process and propagate information, with a focus on Parkinson's disease.
- Examined altered spiking activity and neural synchronization to characterize disease-related network dynamics and temporal symptom patterns.

Georgia Institute of Technology, USA Aug 2023 – Feb 2024
Visiting Researcher (Advisor: Jungho Lewe)

- Developed deep learning models for anomaly detection in high-frequency time-series data, focusing on temporal representations under irregular and noisy conditions.
- Investigated model behavior and temporal patterns in sensor data through data analysis and visualization.

Korea University Mar 2021 – Feb 2023
Graduate Researcher (Advisor: Jun-Geol Baek)

- Developed a distribution-based Transformer framework for time-series missing value imputation (Master's thesis) to improve representation learning from noisy and incomplete sequential data.
- Developed a multiple-encoder framework for multivariate time-series anomaly detection to learn robust temporal representations under varying normality conditions.
- Developed predictive models for Remaining Useful Life (RUL) estimation from high-dimensional vehicle telemetry and operational logs to support maintenance decisions.
- Applied causal inference and statistical analysis to identify critical failure variables and operational vulnerabilities.

Sookmyung Women's University Jul 2019 – Jul 2020
Research Intern (Advisor: Heejoon Chae — Joint Research with Seoul National University Hospital)

- Developed neural network models using multi-omics gene expression data to classify heterogeneous breast cancer subtypes.

Sookmyung Women's University Jun 2020 – Jul 2020
Research Intern (Advisor: Yangjin Kim)

- Applied parametric survival modeling and statistical analysis to COVID-19 clinical records to investigate epidemiological risk factors.

RESEARCH PROJECTS

Cortico-Basal Ganglia Neural Dynamics and Network Modeling 2024 – Present
Korea Institute of Science and Technology (KIST)

- Developing a biologically grounded spiking neural network model to investigate how topographic connectivity shapes information propagation and population dynamics in cortico-basal ganglia circuits.
- Analyzing population activity and neural synchronization to understand how network structure gives rise to emergent neural dynamics in Parkinson's disease.
- Investigating computational mechanisms underlying altered neural activity and their implications for adaptive deep brain stimulation.

AI-Based Deepfake Audio Detection Aug 2026 – Present
DACON Competition Project

- Developing a deepfake audio detection pipeline combining Whisper-based speech transcription with LLM-based analysis to identify AI-generated speech.
- Designing an integrated speech-language analysis framework to improve detection reliability across diverse audio inputs.

PUBLICATIONS

J. Kim, Jun-Geol Baek. A distribution-based transformer framework for time series missing value imputation, Master's Thesis, Korea University, Department of Industrial and Management Engineering, 2023.

J. Ki, **J. Kim**, S. Kim, J. Park, J. Lee, and Y.-J. Kim. Statistical analysis of estimating incubation period distribution and case fatality rate of COVID-19, The Korean Journal of Applied Statistics, 2020.

Joungmin Choi, Jiyoung Lee, **Jieun Kim**, Jihyun Kim, Heejoon Chae. Breast Cancer Subtype Classification Using Multi-omics Data Integration Based on Neural Network, Journal of KIISE, 2020.

PRESENTATIONS

Topographic Cortico-Basal Ganglia Computational Model Reveals Motor and Non-motor Symptom Comorbidity in Parkinson's disease, Society for Neuroscience (SfN), 2025.

Computational Cortico-Basal Ganglia Model Reveals Altered Spiking Dynamics and Synchronization in Parkinson's disease, KSBNS, 2025.

Spiking Neural Network Modeling of Basal Ganglia for Parkinson's disease Prognosis and Adaptive DBS, KSBNS, 2024.

Fuel Cell Vehicle Fuel Efficiency Prediction via Data Refinement and Adversarial Generation, Korean Institute of Industrial Engineers, 2022.

TEACHING AND PROFESSIONAL SERVICE

Corporate Instructor & Programming Assistant Spring 2022 – Dec 2023

- Developed and delivered machine learning and programming curricula for industry practitioners at major enterprise partners, including **Samsung Electronics**, LG Energy Solution, LG Electronics, SK Hynix, and Hyundai Steel.

Teaching Assistant – Korea University Mar 2022 – Dec 2022

- IDS505: Data Science & AI Programming (Spring 2022)
- IDS508: Principles and Applications of Machine Learning (Fall 2022)

Academic Service 2023

- Journal Reviewer: *Journal of Imaging Science and Technology* (2023)
- Book Reviewer: *Insights from Kaggle Medalists* (2023)

AWARDS AND HONORS

Special Prize, Weather Big Data Contest 2022
Grand Prize, Data-based Service Contest for Workplace Safety AI 2021
Third Prize, Contest for Natural and Artificially Generated Data 2021
Best Paper Award, Korean Institute of Information Scientists and Engineers (KIISE) 2020

TECHNICAL SKILLS

Programming Languages: Python (PyTorch, TensorFlow, Brian2), R
Tools & Frameworks: Git, LaTeX, Data Visualization (Tableau, Matplotlib)